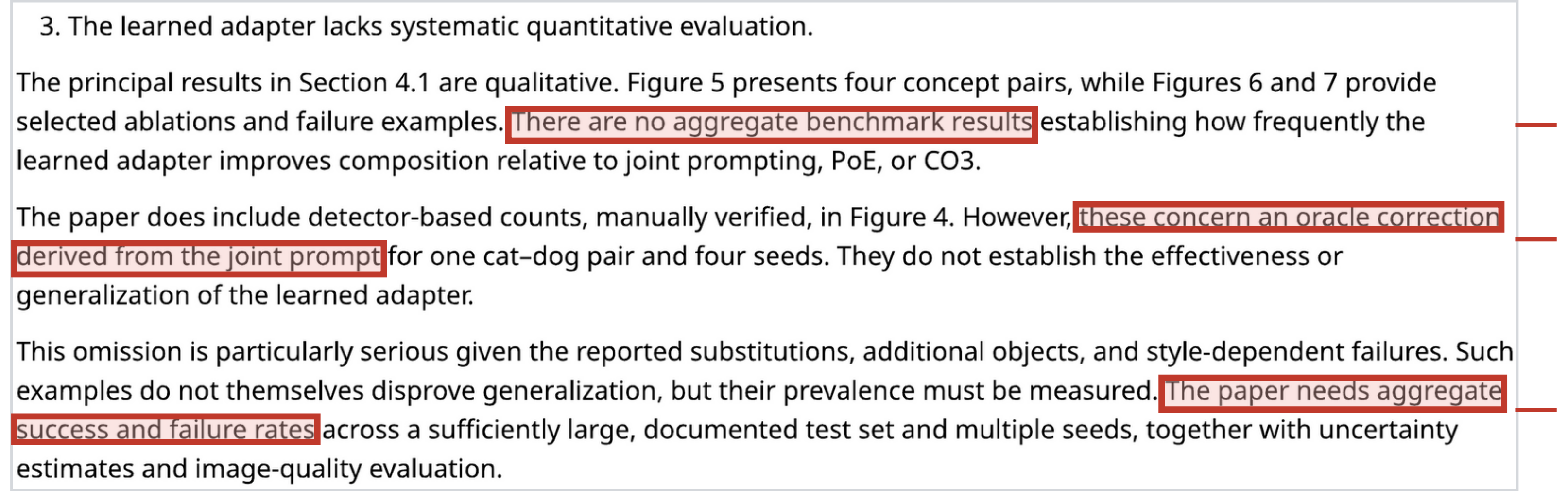
G3: How often the adapter works on unseen pairs

k4nd weakness 3 and question 2, UHz6 weakness 1 and question 2 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 3



3. The learned adapter lacks systematic quantitative evaluation.

The principal results in Section 4.1 are qualitative. Figure 5 presents four concept pairs, while Figures 6 and 7 provide selected ablations and failure examples. There are no aggregate benchmark results establishing how frequently the learned adapter improves composition relative to joint prompting, PoE, or CO3.

The paper does include detector-based counts, manually verified, in Figure 4. However, these concern an oracle correction derived from the joint prompt for one cat–dog pair and four seeds. They do not establish the effectiveness or generalization of the learned adapter.

This omission is particularly serious given the reported substitutions, additional objects, and style-dependent failures. Such examples do not themselves disprove generalization, but their prevalence must be measured. The paper needs aggregate success and failure rates across a sufficiently large, documented test set and multiple seeds, together with uncertainty estimates and image-quality evaluation.

K-W3a

K-W3b

K-W3c

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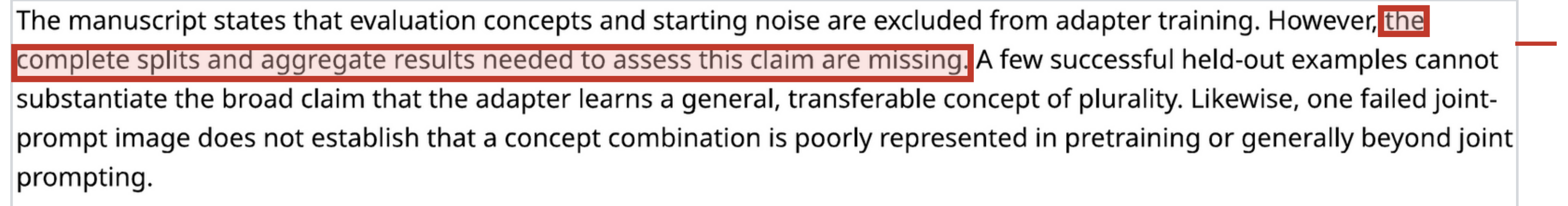
k4nd, weakness 3 (continued)



The manuscript states that evaluation concepts and starting noise are excluded from adapter training. However, the complete splits and aggregate results needed to assess this claim are missing. A few successful held-out examples cannot substantiate the broad claim that the adapter learns a general, transferable concept of plurality. Likewise, one failed joint-prompt image does not establish that a concept combination is poorly represented in pretraining or generally beyond joint prompting.

K-W3d

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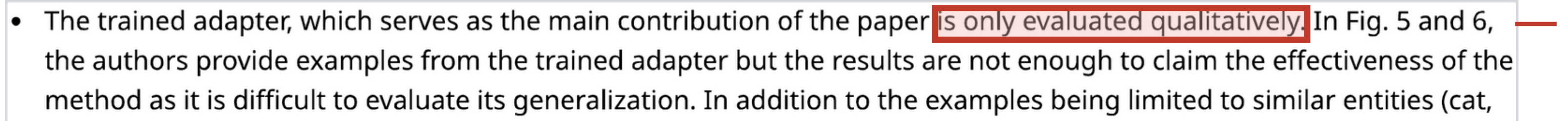
UHz6, weakness 1 and question 2



- The trained adapter, which serves as the main contribution of the paper is only evaluated qualitatively. In Fig. 5 and 6, the authors provide examples from the trained adapter but the results are not enough to claim the effectiveness of the method as it is difficult to evaluate its generalization. In addition to the examples being limited to similar entities (cat,

U-W1a

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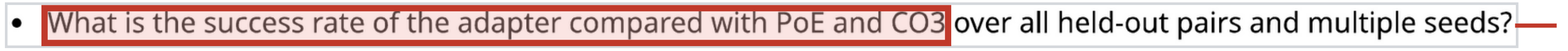
UHz6, question 2



- What is the success rate of the adapter compared with PoE and CO3 over all held-out pairs and multiple seeds?

U-Q2

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these counts

this claim

THE SUBMITTED WEAKNESS

Figure 4, the only counts

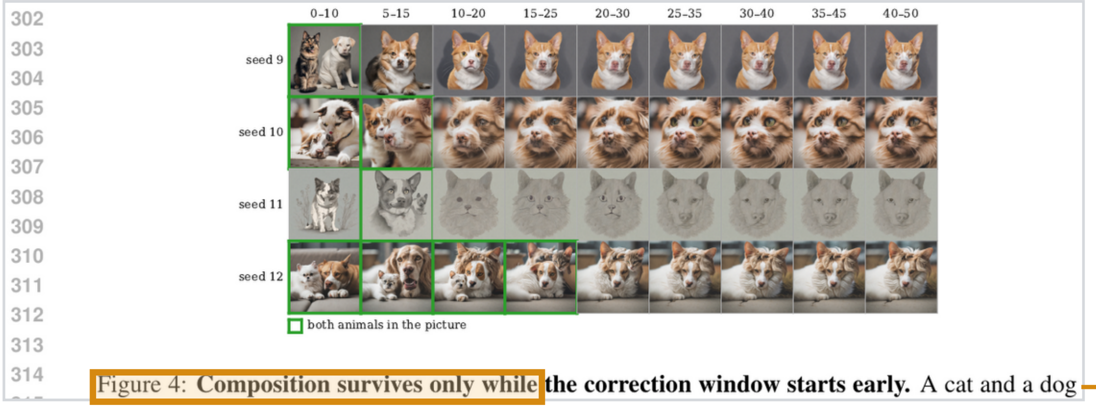
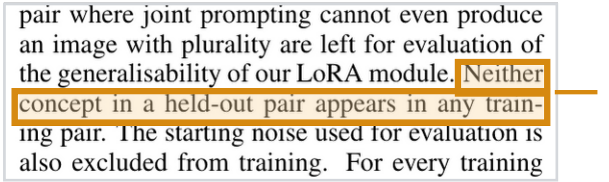


Figure 4: **Composition survives only while the correction window starts early.** A cat and a dog

oracle correction, one pair

56173_Restoring_plurality_in_c.pdf (submitted), page 6

Section 4, the held-out claim



pair where joint prompting cannot even produce an image with plurality are left for evaluation of the generalisability of our LoRA module. Neither concept in a held-out pair appears in any training pair. The starting noise used for evaluation is also excluded from training. For every training

no concept shared?

56173_Restoring_plurality_in_c.pdf (submitted), page 7

learned adapter, 276 prompts

failure rates

protocol stated

THE REVISED ANSWER

Table 1, Attend-and-Excite prompts

every image scored

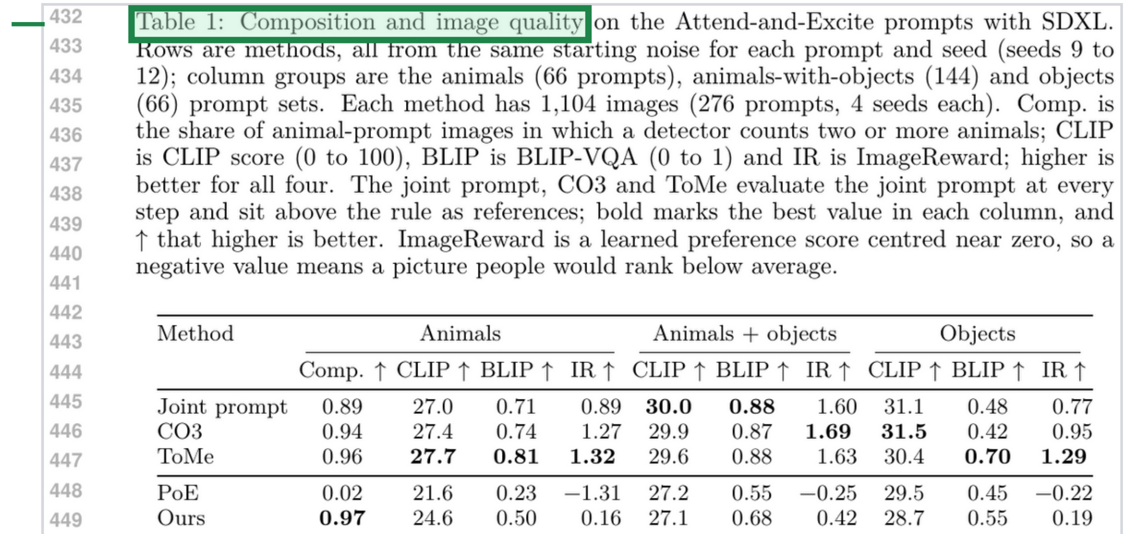


Table 1: Composition and image quality on the Attend-and-Excite prompts with SDXL. Rows are methods, all from the same starting noise for each prompt and seed (seeds 9 to 12); column groups are the animals (66 prompts), animals-with-objects (144) and objects (66) prompt sets. Each method has 1,104 images (276 prompts, 4 seeds each). Comp. is the share of animal-prompt images in which a detector counts two or more animals; CLIP is CLIP score (0 to 100), BLIP is BLIP-VQA (0 to 1) and IR is ImageReward; higher is better for all four. The joint prompt, CO3 and ToMe evaluate the joint prompt at every step and sit above the rule as references; bold marks the best value in each column, and ↑ that higher is better. ImageReward is a learned preference score centred near zero, so a negative value means a picture people would rank below average.

Method	Animals				Animals + objects			Objects		
	Comp. ↑	CLIP ↑	BLIP ↑	IR ↑	CLIP ↑	BLIP ↑	IR ↑	CLIP ↑	BLIP ↑	IR ↑
Joint prompt	0.89	27.0	0.71	0.89	**30.0**	**0.88**	1.60	31.1	0.48	0.77
CO3	0.94	27.4	0.74	1.27	29.9	0.87	**1.69**	**31.5**	0.42	0.95
ToMe	0.96	**27.7**	**0.81**	**1.32**	29.6	0.88	1.63	30.4	**0.70**	**1.29**
PoE	0.02	21.6	0.23	−1.31	27.2	0.55	−0.25	29.5	0.45	−0.22
Ours	**0.97**	24.6	0.50	0.16	27.1	0.68	0.42	28.7	0.55	0.19

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 9

Figure 11, failure types counted

failures counted

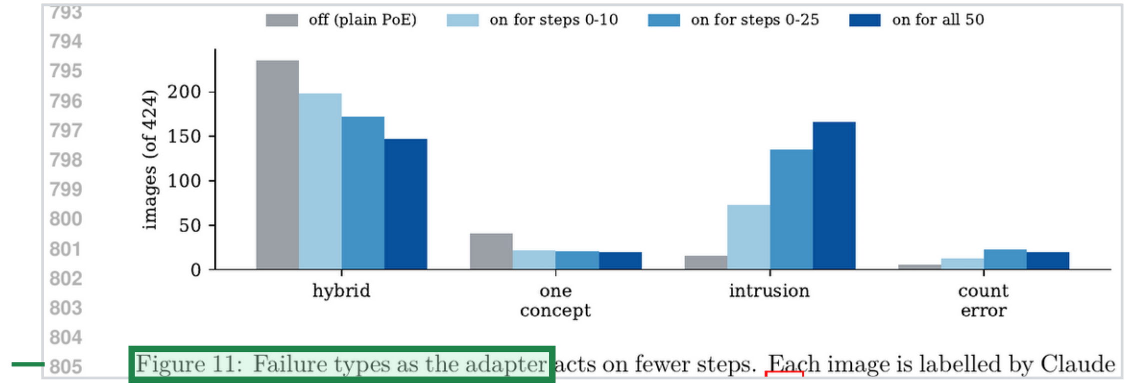
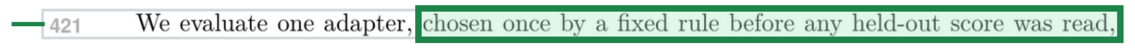


Figure 11: Failure types as the adapter acts on fewer steps. Each image is labelled by Claude

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 15

Section 4.1, the protocol

fixed before scoring



We evaluate one adapter, chosen once by a fixed rule before any held-out score was read,

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W3 and Q2, UHz6 Q2)

answered

W3: no systematic quantitative evaluation of the adapter

We agree, and the revision adds this evaluation. The adapter, the seeds and the success bar were all fixed before any held-out score was read (Section 4.1).

Table 1 scores every method on the 276 Attend-and-Excite prompts (66 animal pairs, 144 animal-object pairs, 66 object pairs) at seeds 9 to 12, with composition measured by BLIP-VQA and image quality by CLIP score and ImageReward.

The training and held-out pairs, with every seed, are released with the code.

Hybrids, missing concepts and additional objects are counted per setting in Figure 11 (Appendix C.2).

Q2: aggregate results over held-out concepts and seeds

Answered under k4nd W3: Table 1, and Figure 11, which counts the failure types the reviewer names.

Q2: success rate against PoE and CO3 over all held-out pairs and seeds

Answered under k4nd W3: Table 1, and Figures 11 and 12 for the held-out pairs at eight seeds.

Changes behind this page: Table 1 scores every method on the 276 Attend-and-Excite prompts, Figure 11 counts failure types on 53 held-out pairs, and Section 4.1 opens with the protocol.
UHz6 weakness 1 also asks for standard benchmarks; that half is on the G4 page. The reply's anonymous code link is still a TODO. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.